

RAHAMOTULLAH MANIK

R&D Assistant Engineer (Mechanical)

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SUMMARY

Mechanical engineer with hands-on R&D experience in CAD design, prototyping and precision manufacturing. Certified SolidWorks Associate (CSWA, 240/240) with a track record spanning military-grade robotics, large-format 3D printing systems, and automated material-handling vehicles — from first sketch through sourcing, fabrication and testing.

SKILLS

- CAD & Engineering: SolidWorks (CSWA Certified), AutoCAD, 2D Mechanical Drafting, 3D Product Modeling & Assemblies
- Industrial Operations: BOM/BOQ, Preventive Maintenance, Machinery Troubleshooting, Quality Control, Calibration & Testing
- Sourcing & Planning: Component Procurement, Vendor Sourcing & Price Negotiation, Project Tracking, Inventory Management
- Collaboration: Cross-functional Teamwork, Technical Documentation, Rapid Problem-Solving, Team Leadership

EXPERIENCE

R&D Assistant Mechanical Engineer

Jan 2024 – Present

Cybernetics Hi-Tech Solutions Ltd., Dhaka

- Designed and developed mechanical components and product assemblies for new and existing R&D projects.
- Prepared 2D/3D CAD models and technical documentation; coordinated with manufacturing for design feasibility.
- Tested and refined prototypes to meet functional and safety standards.
- Managed procurement, vendor sourcing, price negotiation and inventory for project components.

Lab Assistant, Dept. of Nuclear Science & Engineering

Jan 2023 – Dec 2024

Military Institute of Science and Technology, Dhaka

- Supported practical classes in Applied Mechanics, Materials & Metallurgy, Fluid Mechanics, and Thermodynamics & Refrigeration.
- Performed equipment calibration and preventive maintenance; managed lab setup and inventory.

Trainee Engineer

Jul 2022 – Dec 2022

Coca-Cola Beverage Bangladesh Ltd., Mymensingh

- Operated and monitored production line equipment for smooth, efficient beverage manufacturing.
- Assisted preventive maintenance and collaborated on troubleshooting and process improvements.

Part-Time Teacher

Jan 2022 – Jun 2022

Dhaka Polytechnic Institute

- Taught Applied Mechanics and supervised labs in materials, fluid mechanics, and thermodynamics.
- Provided hands-on CAM and machine-operation training; evaluated students through practicals.

KEY PROJECTS

Remotely Operated Vehicle (ROV) — Military-Grade

- Assessed damage and prepared corrected design drawings for workshop fabrication; sourced components to military spec; oversaw precision manufacturing, assembly and QC testing.

Large-Format 3D Printer (500×500 mm bed)

- Designed and built a large-format 3D printer end-to-end — frame, motion system, extruder, heated bed — independently sourcing all components and validating through iterative test prints.

Telescopic Beam for Robotic Arm

- Designed a telescopic beam optimized for compactness and strength-to-weight ratio; CNC-machined parts to tight tolerances; validated load capacity and structural integrity.

Automated Guided Vehicle (AGV)

- Developed an AGV for automated material handling with integrated sensors, motors and control systems for path tracking and obstacle detection.

EDUCATION

- Diploma in Mechanical Technology — Dhaka Polytechnic Institute, 2021 (CGPA 3.71/4.00)
- Secondary School Certificate (Science) — Sreepur Kumaria High School, 2017 (GPA 4.86/5.00)

CERTIFICATION

- Certified SolidWorks Associate (CSWA) — Dassault Systèmes, Score 240/240

LANGUAGES

Bengali (native) · English (professional working proficiency)